

A Lightweight Adaptive Multimodal 3D CNN for ADHD Classification Using Structural MRI and Phenotypic Features

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Abstract. Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder whose diagnosis primarily relies on behavioral and clinical assessment, motivating the investigation of neuroimaging-based computational approaches. However, many deep learning models for neuroimaging employ increasingly complex architectures with substantial parameter and computational requirements. This work presents *ADHDNet*, a lightweight multimodal 3D convolutional neural network for ADHD classification using structural magnetic resonance imaging (sMRI) and phenotypic information. The proposed framework employs depthwise-separable 3D convolutional blocks to reduce model parameters while retaining volumetric feature extraction capability. Three-dimensional efficient channel attention is applied within each encoder block to recalibrate informative feature channels, while an adaptive gated fusion module dynamically combines MRI-derived and phenotypic representations. Experiments were conducted on a multisite subset of the ADHD-200 dataset comprising 497 subjects after excluding a site without ADHD cases. Structural MRI volumes were resized to $64 \times 64 \times 64$, normalized, and combined with five phenotypic features. The proposed model contains only 71,558 trainable parameters (0.0716M). On a stratified held-out validation set, ADHDNet achieved 89.00% accuracy and an AUC of 0.9535, with a sensitivity of 93.33%, specificity of 85.45%, precision of 84.00%, and F1-score of 0.88. Compared with a standard Conv3D multimodal comparator containing 1.376M parameters, ADHDNet reduces the parameter count by approximately 94.8% and its FP32 weight storage from about 5.25 MiB to 0.27 MiB. These results indicate that a compact multimodal 3D architecture can support ADHD classification while substantially reducing model complexity.

Keywords: ADHD · Structural MRI · Multimodal learning · 3D CNN · Depthwise-separable convolution · Efficient channel attention · Lightweight deep learning

1 Introduction

Attention-Deficit/Hyperactivity Disorder (ADHD) is a neurodevelopmental disorder characterized by persistent patterns of inattention, hyperactivity, and im-

pulsivity. Diagnosis primarily relies on behavioral assessment and clinical evaluation, which can be challenging because ADHD symptoms may overlap with those of other developmental and psychiatric conditions. These challenges have motivated the investigation of neuroimaging-based computational approaches for identifying brain patterns associated with ADHD [1]–[3].

Among neuroimaging modalities, structural magnetic resonance imaging (sMRI) provides non-invasive information about brain anatomy and has been widely investigated for computational ADHD analysis [4]. Earlier machine learning studies typically relied on manually derived neuroanatomical measurements, such as gray-matter volume and cortical thickness, together with demographic or personal characteristics. For example, Lohani and Rana [5] combined structural MRI-derived features with personal characteristics and reported a maximum accuracy of 75% using conventional machine learning classifiers on an age-balanced ADHD dataset.

The adoption of deep learning has subsequently enabled direct learning of imaging representations from neuroimaging data, with recent ADHD studies exploring convolutional networks, multimodal architectures, attention mechanisms, and ensemble approaches [6]–[9]. While these approaches can improve representation learning, increasing architectural complexity may also increase the number of trainable parameters and computational requirements. Consequently, parameter efficiency is becoming an important consideration for neuroimaging models intended for resource-constrained environments. Deepika et al. [10] explicitly investigated parameter-efficient ADHD classification using a lightweight attention-driven KAN framework and reported 79.25% accuracy with only a few thousand parameters.

However, relatively limited work has investigated parameter-efficient multimodal architectures that directly process volumetric structural MRI while incorporating phenotypic information. In particular, there remains a need to examine whether volumetric structural information can be modeled using a compact 3D architecture without relying on large conventional convolutional backbones or complex multimodal fusion mechanisms. This study therefore investigates whether a compact 3D architecture can jointly model structural MRI and phenotypic information while substantially reducing parameter requirements.

To address this gap, we propose *ADHDNet*, a lightweight adaptive multimodal 3D CNN designed to jointly exploit volumetric structural MRI and phenotypic information under a constrained parameter budget. The architecture combines depthwise-separable 3D convolution for efficient volumetric feature extraction, 3D efficient channel attention for feature recalibration, and adaptive gated fusion for multimodal integration. The main contributions are as follows:

1. A **lightweight 3D MRI encoder** based on depthwise-separable Conv3D for parameter-efficient volumetric representation learning.
2. Integration of lightweight **3D efficient channel attention (ECA)** within the hierarchical MRI encoder for channel-wise feature recalibration.
3. An **adaptive gated fusion** mechanism for sample-dependent integration of structural MRI and phenotypic representations.

4. An **explicit evaluation of parameter efficiency** against a conventional multimodal Conv3D architecture on a multisite ADHD-200 cohort.

2 Related Work

2.1 MRI-Based ADHD Classification

MRI-based machine learning has been widely investigated for ADHD classification [1]–[3]. Earlier approaches generally relied on manually extracted neuroanatomical or functional connectivity features followed by conventional classifiers. Lohani and Rana [5] extracted gray-matter volume and cortical thickness features from structural MRI, combined them with personal characteristics, and achieved 75% accuracy using conventional classifiers with repeated ten-fold cross-validation. More recent studies have adopted deep learning to learn representations directly from neuroimaging data. CNN-based approaches have been investigated for fMRI-based ADHD classification [6], [11], graph-theoretical features of resting-state networks have been combined with ensemble classifiers [12], graph convolutional networks with multi-head attention have been applied to hybrid-order brain networks [13], and Pedrollo et al. [9] reported CNN ensemble-based classification using resting-state fMRI in a cohort of 623 participants. These studies demonstrate the potential of deep neural networks for learning discriminative neuroimaging representations; however, many focus primarily on predictive performance achieved through increasingly complex architectures.

2.2 Multimodal ADHD Analysis

Phenotypic variables can provide information that is complementary to neuroimaging measurements. Age, sex, medication status, and clinical characteristics may capture information that is not directly represented in image features, and Lohani and Rana [5] provide evidence for the usefulness of combining structural MRI-derived characteristics with personal information. Deepika et al. [14] similarly combined resting-state functional connectivity with phenotypic data. A recent preprint has also explored 3D structural MRI and clinical phenotypic information using adaptive gated multimodal fusion, reporting 87% accuracy on the Peking-1 subset of ADHD-200. In contrast, the present work focuses specifically on reducing model complexity through depthwise-separable 3D convolution and lightweight channel attention while evaluating a multisite cohort.

2.3 Lightweight and Parameter-Efficient ADHD Models

Model complexity is an important consideration when deep learning systems are intended for computationally constrained environments. Deepika et al. [10] explicitly investigated parameter efficiency for ADHD classification and proposed an attention-driven KAN architecture with only a few thousand parameters. Following this direction, the present study treats *parameter efficiency as a primary design objective* rather than a secondary consideration. ADHDNet uses

Table 1. Dataset distribution after site filtering.

Site	ADHD	Control	Total
KKI	22	60	82
NYU	115	96	211
OHSU	36	35	71
Peking_1	24	61	85
NeuroIMAGE	25	23	48
Total	222	275	497

depthwise-separable 3D convolution to reduce the number of learnable parameters while retaining volumetric processing capability.

2.4 Research Gap

Existing studies have demonstrated the value of multimodal information and, separately, of parameter-efficient model design for ADHD classification. However, comparatively few studies combine both directions by processing volumetric structural MRI together with phenotypic information under an explicit parameter budget. ADHDNet is designed to address this combination.

3 Materials and Methods

3.1 Dataset

Experiments were conducted using the preprocessed anatomical component of the publicly available ADHD-200 dataset [15], [16]. The dataset initially contained 575 successfully processed subjects (353 controls and 222 ADHD) distributed across six sites: KKI, NYU, OHSU, Peking_1, Pittsburgh, and NeuroIMAGE. The Pittsburgh site contained 78 control subjects but no ADHD subjects. Since this site does not provide both diagnostic classes required for binary discrimination, it was excluded from model development. After this site-level filtering, the final cohort consisted of 497 subjects (222 ADHD and 275 controls). The distribution after site exclusion is shown in Table 1.

3.2 MRI Preprocessing

The structural MRI data were provided in NIfTI format. To obtain a computationally manageable input representation, each MRI volume was resized to $64 \times 64 \times 64$ voxels using trilinear interpolation. Each resulting volume was independently normalized using min-max normalization,

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \quad (1)$$

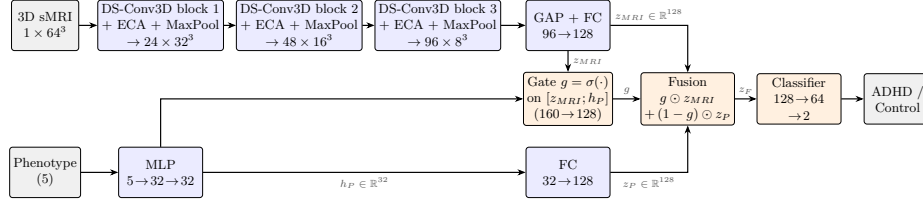


Fig. 1. Overview of ADHDNet as implemented. The sMRI branch consists of three depthwise-separable (DS) Conv3D blocks (24, 48 and 96 channels); each block applies batch normalization, ReLU, 3D efficient channel attention (ECA) and $2 \times 2 \times 2$ max pooling, so channel recalibration is performed at every stage. Global average pooling (GAP) and a linear projection give $z_{MRI} \in \mathbb{R}^{128}$. The phenotypic MLP gives $h_P \in \mathbb{R}^{32}$. The gate is computed from the 160-dimensional concatenation $[z_{MRI}; h_P]$, while h_P is separately projected to $z_P \in \mathbb{R}^{128}$ for gated element-wise fusion.

such that voxel intensities were mapped to the range $[0, 1]$. A relatively simple preprocessing pipeline was adopted to maintain a compact and reproducible experimental setup. No additional skull stripping, atlas-based segmentation, N4 bias correction, or anatomical segmentation was applied.

3.3 Phenotypic Features

Five phenotypic variables were incorporated: age, sex, medication status, site, and ADHD Index. Missing values represented by -999 were converted to missing values before imputation. Continuous variables were normalized, while categorical variables were numerically encoded. The resulting features were passed through a compact multilayer perceptron (Sect. 4.3) to obtain a low-dimensional phenotypic representation complementary to the MRI representation.

4 Proposed ADHDNet

ADHDNet consists of a lightweight 3D MRI encoder, a compact phenotypic encoder, and an adaptive gated fusion module (Fig. 1). A $1 \times 64 \times 64 \times 64$ sMRI volume is processed through three depthwise-separable Conv3D blocks with channel progression $24 \rightarrow 48 \rightarrow 96$, each containing its own 3D ECA module and max pooling, followed by global average pooling and projection to a 128-dimensional MRI representation. In parallel, five phenotypic features are encoded by a $5 \rightarrow 32 \rightarrow 32$ MLP. A gating network computes fusion weights from the concatenated 160-dimensional vector of the MRI and phenotypic representations, the phenotypic representation is projected to 128 dimensions, and the two 128-dimensional representations are combined by gated element-wise fusion and passed to a $128 \rightarrow 64 \rightarrow 2$ classification head. The complete model contains 71,558 trainable parameters. Each component is described below.

4.1 Lightweight 3D MRI Encoder

The imaging branch consists of three depthwise-separable 3D convolutional blocks with channel progression $1 \rightarrow 24 \rightarrow 48 \rightarrow 96$. Each block consists of a depthwise-separable Conv3D operation followed by batch normalization, ReLU activation, 3D efficient channel attention (ECA; Sect. 4.2), and $2 \times 2 \times 2$ max pooling. Channel-wise feature recalibration is therefore performed at each hierarchical feature-extraction stage, and the spatial resolution is halved by each block ($64^3 \rightarrow 32^3 \rightarrow 16^3 \rightarrow 8^3$). Depthwise-separable convolution [17], [18] decomposes a conventional convolution into (i) a depthwise spatial convolution and (ii) a pointwise $1 \times 1 \times 1$ convolution. For a conventional 3D convolution with kernel size K , the parameter count is approximately

$$P_{\text{conv}} = K^3 C_{in} C_{out}, \quad (2)$$

whereas depthwise-separable convolution requires approximately

$$P_{\text{DS}} = K^3 C_{in} + C_{in} C_{out}. \quad (3)$$

Since the second expression can be considerably smaller when C_{out} is large, depthwise-separable convolution provides a mechanism for reducing the number of trainable parameters. After the third block, adaptive global average pooling of the $96 \times 8 \times 8 \times 8$ feature map produces a 96-dimensional representation, which is projected to a 128-dimensional MRI embedding.

4.2 Efficient Channel Attention

Within each encoder block, 3D ECA recalibrates the channels produced by the convolution before max pooling, giving three ECA modules in total. To improve feature recalibration without introducing a large parameter overhead, a three-dimensional extension of efficient channel attention (ECA) [19] is incorporated. The attention operation estimates channel importance from globally pooled feature responses and applies channel-wise weights to the volumetric feature maps:

$$F_{att} = A(F) \odot F, \quad (4)$$

where F represents the feature tensor and $A(F)$ the learned channel attention weights. This allows the network to emphasize informative feature channels while maintaining the lightweight nature of the architecture.

4.3 Phenotypic Encoder

In parallel, the phenotypic branch encodes the five tabular variables into a compact representation. The phenotype encoder is a small multilayer perceptron with dimensions $5 \rightarrow 32 \rightarrow 32$. ReLU activation, batch normalization [20], and dropout [21] are used to regularize the representation. The resulting 32-dimensional phenotype representation is used directly by the gating network and is separately projected to 128 dimensions for element-wise fusion (Sect. 4.4).

4.4 Adaptive Gated Fusion

The two modality-specific embeddings are subsequently integrated using adaptive gated fusion. Simple concatenation treats both modalities equally at the fusion stage; instead, ADHDNet employs a learnable gating mechanism, in the spirit of gated multimodal units [22]. The MRI branch produces a 128-dimensional representation $z_{MRI} \in \mathbb{R}^{128}$, while the phenotypic MLP produces a 32-dimensional representation $h_P \in \mathbb{R}^{32}$. These representations are concatenated to form a 160-dimensional vector $z = [z_{MRI}; h_P] \in \mathbb{R}^{160}$, from which the gating network generates

$$g = \sigma(W_2 \delta(W_1 z)) \in \mathbb{R}^{128}, \quad (5)$$

where σ denotes the sigmoid function and δ a nonlinear activation. The phenotypic representation is independently projected to 128 dimensions, $z_P = \text{Proj}(h_P) \in \mathbb{R}^{128}$, and the fused representation is computed by gated element-wise fusion:

$$z_F = g \odot z_{MRI} + (1 - g) \odot z_P. \quad (6)$$

Thus, the model can dynamically determine the relative contribution of the imaging and phenotypic representations for each sample.

4.5 Classification Head

The fused 128-dimensional representation is passed through a compact classifier with dimensions $128 \rightarrow 64 \rightarrow 2$, whose two output units correspond to the ADHD and control classes. Softmax probabilities are used to obtain the predicted class.

5 Model Complexity

Parameter efficiency is a central design consideration of ADHDNet. The proposed model contains 71,558 trainable parameters ($\approx 0.0716\text{M}$). For comparison, a standard Conv3D multimodal architecture implemented under the same experimental framework contains 1,376,080 parameters ($\approx 1.376\text{M}$). The relative reduction is

$$\frac{1,376,080 - 71,558}{1,376,080} \times 100 \approx 94.8\%. \quad (7)$$

Stored as 32-bit floating-point values, the trained weights of ADHDNet occupy $71,558 \times 4 = 286,232$ bytes (≈ 0.27 MiB), compared with $1,376,080 \times 4 = 5,504,320$ bytes (≈ 5.25 MiB) for the comparator. These figures describe FP32 weight storage only, not runtime memory usage. The efficiency results are visualized in Fig. 2(b) (Sect. 7.4).

6 Experimental Setup

6.1 Training Strategy

The network was trained in PyTorch [23] using the AdamW optimizer [24] with a learning rate of 1×10^{-4} , weight decay of 1×10^{-4} , batch size of 8, a maximum of 150 epochs, early stopping with patience 15, label smoothing [25] of 0.1, gradient clipping at 1.0, and cosine annealing learning-rate scheduling [26]. Weighted cross-entropy loss was used to account for class imbalance. The random seed was fixed to 42; Python, NumPy, and PyTorch random states were seeded, and deterministic CUDA settings were used where applicable.

6.2 Experimental Protocol

The final dataset was divided using a stratified 80:20 train-validation split, giving 397 training subjects and 100 held-out validation subjects (45 ADHD and 55 controls). Normalization statistics were calculated from the training data and applied to the validation data. Importantly, the validation set was used for early stopping and model selection; therefore, the reported results should be interpreted as *held-out validation performance* rather than independent external test performance.

6.3 Evaluation Metrics

With ADHD as the positive class, the model was evaluated using

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \quad \text{Sensitivity} = \frac{TP}{TP + FN}, \quad (8)$$

$$\text{Specificity} = \frac{TN}{TN + FP}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad (9)$$

$$F_1 = 2 \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (10)$$

Receiver operating characteristic (ROC) analysis was also performed and the area under the ROC curve (AUC) was calculated.

7 Results

7.1 Classification Performance

Table 2 summarizes the classification performance of ADHDNet on the held-out validation set. ADHDNet achieved an accuracy of 89.00% and an AUC of 0.9535, with a sensitivity of 93.33%, specificity of 85.45%, precision of 84.00%, NPV of 94.00%, and F1-score of 0.88.

Table 2. Classification performance of ADHDNet on the held-out validation set ($n = 100$; 45 ADHD, 55 controls). ADHD is the positive class.

Accuracy	AUC	Sensitivity	Specificity	Precision	NPV	F1-score
89.00%	0.9535	93.33%	85.45%	84.00%	94.00%	0.88

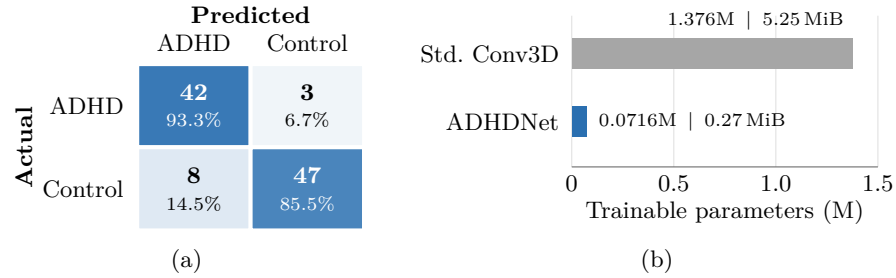


Fig. 2. (a) Confusion matrix of ADHDNet on the held-out validation set ($n = 100$); cell shading and percentages are normalized by the actual class. (b) Trainable parameters of ADHDNet and the standard Conv3D multimodal comparator, a 94.8% reduction; labels give the corresponding FP32 weight storage (parameters $\times$ 4 bytes), not runtime memory usage.

7.2 Per-Class Error Analysis

The confusion matrix in Fig. 2(a) shows how the errors are distributed between the two classes. ADHDNet correctly classified 42 of 45 ADHD subjects and 47 of 55 controls. Of the 11 misclassifications, 8 were controls predicted as ADHD and 3 were ADHD subjects predicted as controls. The model therefore errs more often towards false positives than false negatives, which is reflected in the higher sensitivity (93.33%) than specificity (85.45%) and in the higher NPV (94.00%) than precision (84.00%). In a screening-support setting, this asymmetry means that a negative prediction was more reliable than a positive one on this validation cohort.

7.3 ROC Analysis

The model achieved an AUC of 0.9535 on the held-out validation cohort. However, because this validation set was used for early stopping and model selection, the result should not be interpreted as an independent external validation estimate.

7.4 Computational Efficiency

Because parameter efficiency is a primary design objective of ADHDNet, model complexity was explicitly evaluated against a standard Conv3D multimodal comparator implemented under the same experimental framework (Fig. 2(b)).

ADHDNet contains 0.0716M trainable parameters compared with 1.376M for the comparator, corresponding to a 94.8% reduction in parameter count (a factor of 19.2). Correspondingly, its FP32 weight storage falls from approximately 5.25 MiB to 0.27 MiB. These results show that multimodal 3D MRI analysis does not necessarily require a large number of trainable parameters.

7.5 Comparison with Existing Studies

Existing studies illustrate several directions in ADHD classification, including conventional machine learning using structural MRI-derived features, unimodal deep learning and graph-based models of functional MRI, multimodal approaches, and lightweight models (Table 3). Among multimodal studies, Lohani and Rana [5] reported 75% accuracy using structural MRI-derived features and personal characteristics with conventional classifiers, while Deepika et al. [14] reported 77.59% using resting-state functional connectivity and phenotypic information. Among unimodal studies, Gülhan and Özmen [11] applied a 3D CNN to regional fMRI analysis on ADHD-200 and reported a best per-site accuracy of 76.19%, Zamanzadeh et al. [12] reported 74.30% using graph features of audio-visual integration networks with an EasyEnsemble classifier, and Wu et al. [13] reported 77.95% using the HAGCN graph convolutional framework. More recently, parameter-efficient learning has been investigated using attention-driven KANs, with Deepika et al. [10] reporting 79.25% accuracy using a model with only a few thousand parameters.

These studies provide useful evidence for both multimodal learning and parameter-efficient modeling, but their imaging modalities, datasets, and evaluation protocols differ from those used in the present study; the reported accuracies are therefore not directly comparable. ADHDNet instead focuses on combining volumetric structural MRI and phenotypic information within a compact 3D CNN architecture.

8 Discussion

The results demonstrate the feasibility of using a compact 3D CNN to jointly model structural MRI and phenotypic information for ADHD classification. ADHDNet contains only 71,558 trainable parameters, corresponding to a 94.8% reduction relative to the standard Conv3D comparator under the same experimental framework, while achieving an AUC of 0.9535 on the held-out validation set.

The parameter reduction is primarily achieved through depthwise-separable 3D convolution, while 3D ECA provides channel-wise feature recalibration with limited additional parameter overhead. The adaptive gating mechanism further allows the model to integrate MRI and phenotypic representations through sample-dependent weighting rather than simple concatenation. Together, these choices place ADHDNet within the emerging direction of lightweight ADHD

Table 3. Comparison with selected recent ADHD studies. Accuracies are not directly comparable due to differing modalities, datasets, and protocols.

Study	Year	Modality	Model	Multi-modal	Params	Acc. (%)
Lohani & Rana [5]	2023	sMRI + personal characteristics	SVM/RF/KNN	✓	NR	75.0
Gülhan & Özmen [11]	2024	fMRI	3D CNN	–	NR	76.19 [†]
Deepika et al. [14]	2024	rs-fMRI + phenotypic	Multimodality model	✓	NR	77.59
Zamanzadeh et al. [12]	2024	rs-fMRI	EEC	–	NR	74.30
Wu et al. [13]	2025	rs-fMRI	HAGCN	–	NR	77.95
Deepika et al. [10]	2025	MRI-derived features	Lightweight attention-KAN	✓*	Few thousand*	79.25
ADHDNet (ours)	2026	3D sMRI + phenotype	DS-Conv3D + ECA + gated fusion	✓	0.0716M	89.0

NR: not reported; –: single modality. [†]Best per-site accuracy on ADHD-200 (NeuroIMAGE 76.19%, NYU 69.92%, PU 70.77%). *As described by the authors; exact parameter count not stated. EEC: EasyEnsemble classifier.

models [10], with a specific focus on volumetric structural MRI and phenotypic multimodal learning.

It is important, however, not to interpret the 89% accuracy as evidence that ADHDNet is universally superior to previously published approaches, since existing studies use different datasets, modalities, preprocessing procedures, class definitions, and validation protocols. The main contribution is therefore a *parameter-efficient multimodal 3D architecture that combines depthwise-separable volumetric convolution and lightweight channel attention with adaptive fusion of structural MRI and phenotypic representations*, rather than the highest reported ADHD classification accuracy.

8.1 Limitations and Future Work

Several limitations should be considered. First, the present study uses a single held-out validation split rather than an independent external test cohort; the reported results should therefore not be interpreted as evidence of external generalization. Second, although the dataset contains multiple acquisition sites, the current experiment does not explicitly evaluate cross-site generalization. Third, the simplified preprocessing pipeline, without skull stripping, registration, or atlas-based segmentation, may affect generalization to data acquired under different protocols. Fourth, the current work focuses on binary ADHD-versus-control classification and does not investigate ADHD subtypes. Fifth, component-wise ablation of the ECA modules, the phenotypic branch, and the adaptive fusion mechanism remains to be investigated in future experiments.

Finally, the lightweight claim is currently supported by parameter count and FP32 weight storage; future evaluation should additionally report FLOPs, inference latency, and peak memory consumption under a standardized hardware and software environment.

9 Conclusion

This paper presented ADHDNet, a lightweight multimodal 3D CNN for ADHD classification using structural MRI and phenotypic information. The architecture combines depthwise-separable 3D convolution, efficient channel attention, a compact phenotypic encoder, and adaptive gated fusion. On a multisite subset of ADHD-200 comprising 497 subjects, ADHDNet achieved 89.00% accuracy and an AUC of 0.9535 on a held-out validation set, while using only 71,558 trainable parameters. This corresponds to a 94.8% reduction in parameter count relative to the standard Conv3D comparator implemented under the same experimental framework. These results support the feasibility of parameter-efficient multimodal 3D modeling for ADHD classification. Future work will investigate component-wise ablation, standardized computational profiling, cross-site evaluation, and independent external validation.

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